

R E V I E W

Imaging of adrenal masses: From incidental detection to clinical decision-making. A radiologist's evidence-based vade mecum

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ABSTRACT

Background and aim: The widespread use of cross-sectional imaging has led to an increasing detection of adrenal masses as incidental findings. While most lesions are benign and non-functioning, a clinically relevant proportion may be hormonally active or malignant. Accurate imaging characterization is essential to guide management. This review provides an imaging-focused, evidence-based overview of adrenal mass evaluation, emphasizing radiological decision-making.

Methods: A narrative review of contemporary literature was conducted, focusing on the role of ultrasound, computed tomography (CT), and magnetic resonance imaging (MRI) in adrenal lesion assessment. Validated imaging criteria, diagnostic accuracy, and modality-specific limitations were analyzed, integrating clinical and biochemical considerations.

Results: Characteristic imaging features of common adrenal lesions, including adenomas, myelolipomas, and pheochromocytomas, are described. Less frequent but clinically significant entities are discussed. Strengths and limitations of imaging techniques are outlined, and practical imaging-based diagnostic strategies are proposed.

Conclusions: A structured, multiparametric imaging approach allows confident characterization of most adrenal masses and supports appropriate multidisciplinary management. Awareness of imaging limitations is essential to avoid misclassification and unnecessary investigations. (www.actabiomedica.it)

Key words: adrenal incidentaloma, adrenal tumors, computed tomography, magnetic resonance imaging, ultrasonography



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Introduction

Over recent decades, the detection of adrenal masses has risen dramatically, reflecting both the increasing use of cross-sectional imaging and the improved spatial and contrast resolution of modern CT and MRI scanners (1). Population-based studies estimate the prevalence of adrenal incidentalomas at approximately 3–5%, with higher rates observed in older populations. Most adrenal masses are benign and clinically silent; however, a minority represent hormonally active tumors or malignancies, such as metastatic deposits or primary adrenocortical carcinoma. Consequently, a critical task in clinical practice is to distinguish lesions requiring further evaluation from those that can be safely monitored conservatively (2,3).

Imaging is central to this process. Radiological findings often constitute the first indication of lesion type, influencing subsequent biochemical work-up, follow-up strategy, and therapeutic decisions. For this reason, radiologists must not only recognize the imaging features of adrenal lesions but also appreciate their clinical significance within a structured diagnostic framework (4).

Anatomical and biological considerations

The adrenal glands are paired retroperitoneal organs composed of two embryologically distinct and functionally different components: the cortex and the medulla. This distinction underlies the variable biological behavior and imaging characteristics of adrenal lesions (5,6). The adrenal cortex is responsible for steroid hormone production and is the origin of adenomas, adrenocortical carcinomas, and oncocytic neoplasms. Cortical lesions often contain intracellular lipid, a feature that forms the basis of diagnostic imaging techniques, particularly unenhanced CT attenuation measurement and chemical shift MRI. In contrast, the adrenal medulla, derived from neural crest cells, is the site of pheochromocytomas. These tumors are typically hypervascular and may display heterogeneous internal architecture due to hemorrhage, necrosis, or cystic degeneration, accounting for their variable imaging appearance.

Clinical context and definition of adrenal incidentaloma

An adrenal incidentaloma is defined as a mass identified incidentally during imaging performed for indications unrelated to suspected adrenal disease. Lesions discovered during staging or follow-up of known malignancy, or during dedicated endocrine evaluation, are excluded from this definition. Upon detection, two key questions arise: whether the lesion is malignant and whether it is hormonally active. While definitive assessment relies on integration of imaging, biochemical testing, and clinical evaluation, imaging often provides the earliest and most influential information. Radiologists must therefore be familiar with both the imaging characteristics and the clinical implications of adrenal lesions to guide appropriate management (2–6).

Imaging modalities

Ultrasound

Ultrasound plays a limited role in adult adrenal imaging due to the deep retroperitoneal location of the glands and frequent interference from bowel gas (7,8). Adrenal lesions may occasionally be detected, particularly in slender patients, and ultrasound can provide initial information regarding lesion size, contour, echogenicity, and vascularity. However, the sensitivity and specificity of conventional ultrasound are insufficient for confident lesion characterization, and suspicious findings should prompt further evaluation with cross-sectional imaging such as CT or MRI (Figure 1).

Contrast-Enhanced Ultrasound (CEUS) can provide additional, functionally relevant information by evaluating the microvascular perfusion of adrenal lesions in real time. CEUS allows differentiation between highly vascular tumors, such as pheochromocytomas, and lesions with low perfusion, such as myelolipomas or adenomas, based on their enhancement patterns. Typical benign adenomas demonstrate mild, homogeneous enhancement with rapid wash-in and wash-out, whereas hypervascular or malignant lesions often display heterogeneous enhancement, delayed wash-out, or irregular vascular distribution. CEUS is particularly advantageous in patients who



Figure 1. Transabdominal ultrasound of the right adrenal gland demonstrates a previously known adenoma, well-circumscribed and homogeneous, measuring approximately 15 × 12 mm. The sonographic appearance, while non-specific, illustrates the occasional detectability of adrenal adenomas with ultrasound and highlights the need for further assessment with CT or MRI.

cannot undergo CT or MRI due to contraindications (e.g., renal insufficiency or contrast allergy), and it can guide interventional procedures or follow-up assessment in selected cases.

Despite these benefits, CEUS has limitations: the adrenal glands' deep retroperitoneal location can restrict acoustic access, large body habitus or overlying bowel gas can degrade image quality, and CEUS cannot reliably characterize all lesion subtypes, particularly small or complex masses. Therefore, CEUS should be considered an adjunct tool rather than a standalone diagnostic modality, integrated within a structured, multimodal imaging algorithm to maximize diagnostic confidence and guide clinical management.

Computed tomography

CT is the reference standard for adrenal imaging and is often the first modality to identify adrenal masses (9,10).

Unenhanced CT permits attenuation measurement, a critical step in lesion characterization. Adenomas with intracellular lipid typically display attenuation ≤ 10 Hounsfield units (HU), allowing confident non-invasive diagnosis in most cases. For lesions with higher attenuation, contrast-enhanced CT with delayed imaging permits calculation of absolute and relative contrast washout. Adenomas typically

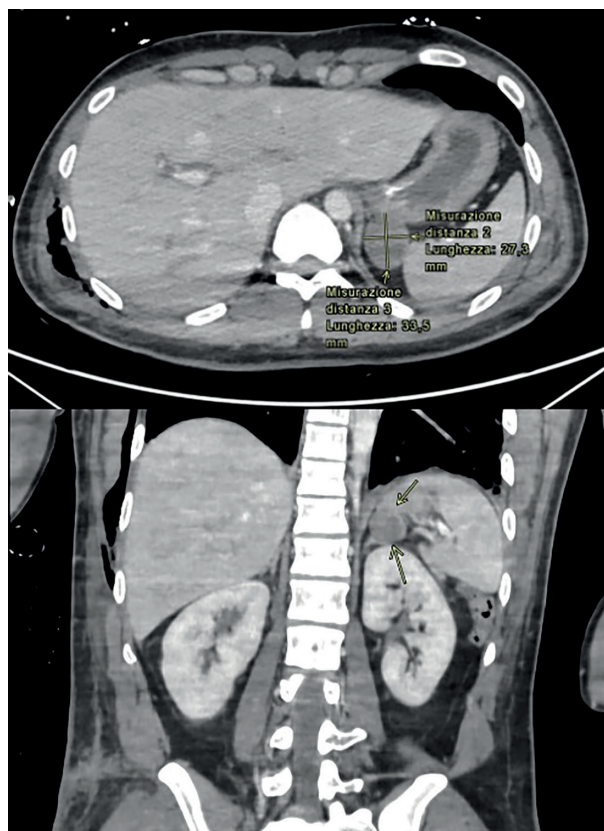


Figure 2. Unenhanced CT of the left adrenal gland shows a 12 × 10 mm lipid-rich adenoma, well-circumscribed and homogeneous, with low attenuation confirming intracellular lipid. CT provides definitive characterization, differentiating adenomas from malignant or indeterminate lesions.

demonstrate rapid washout due to their vascular and interstitial properties, while malignancies often show slow or heterogeneous washout. CT also allows precise assessment of lesion size, morphology, margins, and interval growth, parameters essential for risk stratification and follow-up planning (Figure 2).

Contrast-enhanced CT and washout analysis

For lesions with unenhanced attenuation values greater than 10 HU, contrast-enhanced CT with delayed imaging enables calculation of absolute and relative contrast washout. Rapid contrast washout reflects the vascular and interstitial properties of adenomatous tissue and has been extensively validated as a reliable diagnostic criterion.

CT also allows accurate assessment of lesion size, morphology, margins, and interval growth, which are

essential parameters for risk stratification and follow-up planning.

Magnetic resonance imaging

MRI is particularly useful when CT findings are indeterminate or when radiation exposure should be minimised (11).

Chemical Shift Imaging (CSI) in adrenal mass evaluation

Chemical Shift Imaging (CSI) represents a cornerstone in MRI characterization of adrenal lesions, exploiting differences in resonance frequency between fat and water protons. Adenomas, particularly lipid-rich variants, contain abundant intracellular lipid, which produces a characteristic signal drop on out-of-phase sequences compared with in-phase images. This phenomenon allows reliable differentiation from non-adenomatous lesions, including pheochromocytomas, metastases, and adrenal carcinomas, which typically lack significant intracellular lipid.

Multiple studies have demonstrated the high sensitivity and specificity of CSI, with reported sensitivity for detecting lipid-rich adenomas ranging from 85% to 100% and specificity between 90% and 98% (2). CSI is particularly valuable in cases where CT attenuation is borderline (>10 HU) or when the lesion is small and washout CT is inconclusive. Furthermore, CSI offers the advantage of radiation-free evaluation, making it suitable for follow-up in younger patients or in those requiring repeated imaging.

Despite its strengths, CSI has limitations. Signal loss may be incomplete in lipid-poor adenomas; large lesions with hemorrhage, necrosis, or cystic degeneration may produce heterogeneous results, and very small lesions (<1 cm) may be difficult to characterize reliably. Therefore, CSI should be interpreted in the context of multiparametric MRI, combining T1/T2-weighted sequences, fat-saturation, diffusion-weighted imaging, and, when indicated, contrast-enhanced imaging, to provide a comprehensive non-invasive assessment of adrenal masses.

In clinical practice, CSI serves as a powerful adjunct to CT, refining diagnostic confidence, guiding management decisions, and reducing unnecessary

interventions or biopsies. Its integration into structured imaging algorithms is endorsed by contemporary guidelines (ESE/ENSAT 2016–2023), highlighting its pivotal role in the modern radiological evaluation of adrenal incidentalomas.

Multiparametric MRI

Conventional T1- and T2-weighted imaging, diffusion-weighted imaging, and contrast-enhanced sequences provide complementary information regarding tissue composition, cellularity, and vascularity. MRI is particularly helpful in the evaluation of pheochromocytomas and selected indeterminate adrenal lesions (12–15).

Imaging features of common adrenal lesions

Adrenal adenomas

Adrenal adenomas are the most frequently encountered adrenal tumors in imaging practice. They are commonly detected incidentally and are usually non-functioning, although autonomous hormone secretion may be present (Figure 3).

Imaging features typically include a well-defined and homogeneous appearance. Lipid-rich adenomas show low attenuation on unenhanced CT, whereas lipid-poor adenomas require further characterisation using washout CT or MRI (16–20). Chemical shift MRI demonstrates high sensitivity for detecting intracellular lipid, although overlap with malignant lesions may occur in atypical cases (Figure 4).

Adrenal myelolipoma

Myelolipomas are benign, non-functioning tumors composed of mature adipose tissue and hematopoietic elements (21–22). Their imaging appearance is often characteristic.

On CT, myelolipomas contain macroscopic fat with negative attenuation values, frequently intermixed with soft-tissue components (Figure 5).

MRI confirms fat through high signal intensity on T1-weighted images and signal suppression on



Figure 3. Transabdominal ultrasound evaluation of the left adrenal lodge demonstrates a previously known adrenal cortical adenoma, visualized as a well-circumscribed, solid, and homogeneous lesion measuring approximately 15×12 mm. Although the sonographic appearance is inherently non-specific and does not allow definitive tissue characterization, this finding illustrates the occasional detectability of small adrenal adenomas on ultrasound and emphasizes the ancillary role of sonography in adrenal imaging, with definitive lesion classification requiring correlation with cross-sectional imaging and clinical-biochemical assessment.

fat-saturated sequences (23). Large lesions may be associated with hemorrhage or mass effect, potentially influencing management (Figures 6, 7).

This case illustrates the strength of multiparametric MRI in evaluating adrenal myelolipomas, providing complementary tissue characterization, differentiating fat-containing benign lesions from lipid-poor adenomas or malignant masses, and supporting non-invasive diagnosis without biopsy. The combination of fat-sensitive sequences, chemical shift imaging, and



Figure 4. Unenhanced computed tomography of the abdomen shows a right adrenal cortical adenoma measuring approximately 12×10 mm, appearing as a well-circumscribed, homogeneous lesion. The lesion demonstrates low unenhanced attenuation values (≤ 10 HU), indicative of intracellular lipid content, a hallmark of benign adrenal adenomas. This case illustrates the fundamental role of CT in adrenal imaging, allowing radiologists to confidently distinguish lipid-rich adenomas from malignant or indeterminate masses, guide clinical management, and determine the need—or lack thereof—for further follow-up or invasive procedures.

diffusion assessment makes MRI a powerful tool for confident characterization of adrenal myelolipomas.

Pheochromocytoma are clinically significant tumors with variable imaging appearance. Ultrasound

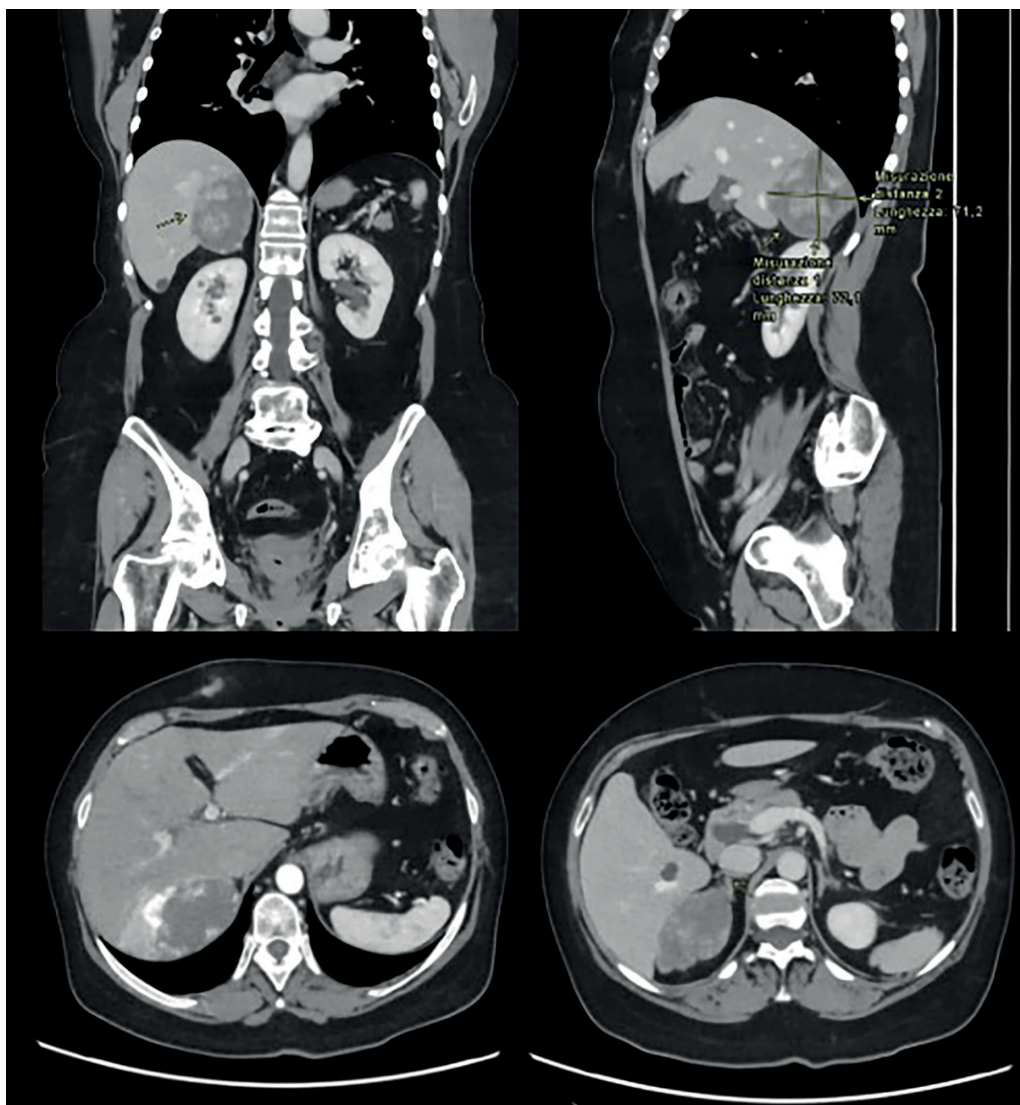


Figure 5. Unenhanced computed tomography of the abdomen demonstrates a right adrenal myelolipoma measuring approximately 71 × 72 mm. The lesion is well-circumscribed and heterogeneous, containing macroscopic fat interspersed with soft-tissue elements, which is characteristic of myelolipomas. Myelolipomas are benign, non-functioning adrenal tumors composed of mature adipose tissue and hematopoietic elements. On CT, the presence of negative attenuation values within the lesion confirms fat content, allowing confident diagnosis without the need for biopsy. This example highlights the pathognomonic imaging appearance of adrenal myelolipomas, illustrating the capability of cross-sectional imaging to differentiate benign fat-containing lesions from other adrenal masses and guide appropriate clinical management, particularly in large lesions that may cause mass effect or hemorrhage.

may show a large, round, well-circumscribed, heterogeneous mass with internal vascularity (24-26). CT demonstrates higher unenhanced attenuation than adenomas and avid, heterogeneous post-contrast enhancement. MRI often shows a high T2 signal and variable enhancement. No single imaging feature is

pathognomonic; biochemical correlation is essential (Figure 8)

On CT, pheochromocytomas typically demonstrate attenuation values higher than those of adenomas and exhibit avid enhancement after contrast administration (Figure 9)

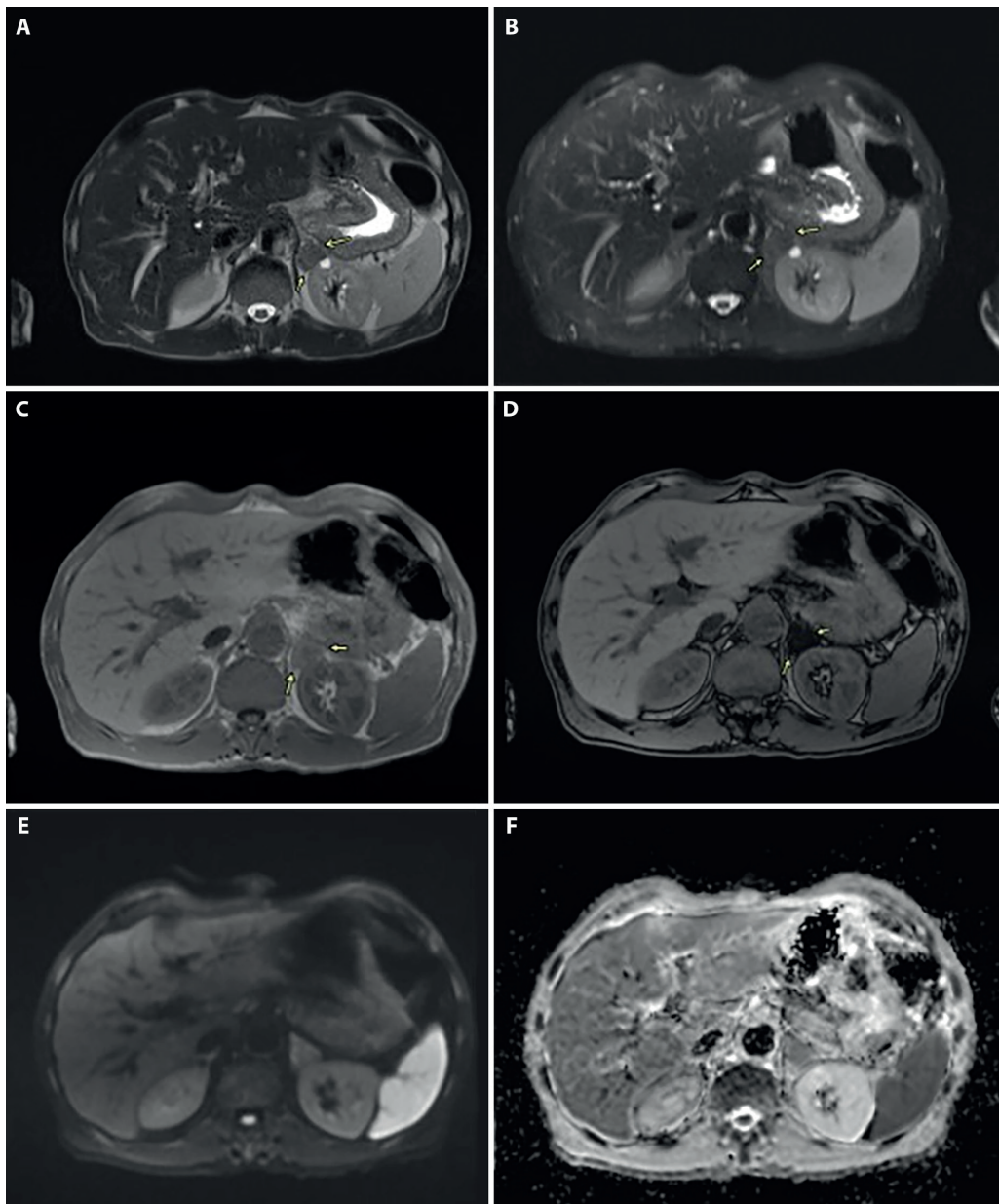


Figure 6. Axial magnetic resonance imaging of a left adrenal myelolipoma demonstrates its characteristic appearance across multiple sequences: A) T2-weighted – the lesion is heterogeneous, with hyperintense areas corresponding to fatty components and intermediate signal reflecting soft-tissue elements. B) T2 FAT SAT – signal from fat is selectively suppressed, highlighting the soft-tissue components and confirming the presence of macroscopic fat. C) In-phase (IN) and D) Out-of-phase (OPP) – chemical shift imaging reveals signal dropout on out-of-phase images, reflecting intracellular lipid, which aids in differentiation from other adrenal lesions. E) Diffusion-weighted imaging (DWI) – the lesion shows no significant diffusion restriction, supporting its benign nature. F) Apparent diffusion coefficient (ADC) map – demonstrates high ADC values, consistent with low cellularity and confirming benign behavior.

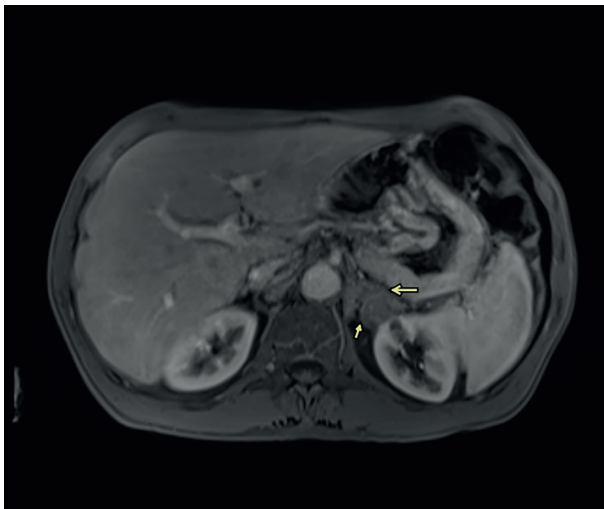


Figure 7. Axial post-contrast T1-weighted MRI of the left adrenal myelolipoma demonstrates selective enhancement of the soft-tissue elements within the lesion, while the macroscopic fat remains non-enhancing. This contrast pattern reinforces the benign diagnosis, distinguishing myelolipoma from adrenal cortical carcinoma or metastasis, which typically exhibit more pronounced and homogeneous enhancement. When interpreted alongside fat-sensitive and chemical shift sequences, contrast-enhanced imaging provides complementary confirmation of tissue composition, guides clinical decision-making, and may help assess lesion vascularity in large or symptomatic masses..

On MRI, they often exhibit high signal intensity on T2-weighted images, although this finding is not universal (27,28).

As no single imaging feature is pathognomonic, pheochromocytoma should be included in the differential diagnosis of indeterminate adrenal masses, particularly before invasive procedures.

Malignant and uncommon adrenal lesions

Adrenal metastases

Metastases are the most common malignant adrenal lesions and are frequently bilateral. Imaging features may overlap with those of lipid-poor adenomas, highlighting the importance of clinical history and prior imaging for correct interpretation (17, 18).

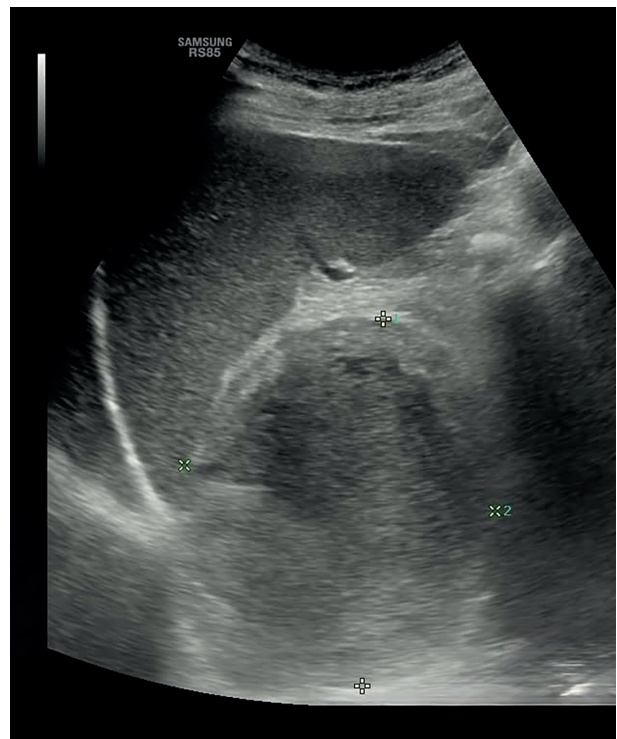


Figure 8. Transabdominal ultrasound of the left adrenal gland demonstrates a large, round, well-circumscribed mass consistent with a pheochromocytoma. The lesion appears heterogeneous with mixed echogenicity and internal vascularity on Doppler imaging, reflecting the tumor's hypervascular and often complex architecture. Although ultrasound is limited in sensitivity for adrenal lesions due to depth and bowel gas, this case illustrates the sonographic appearance of a clinically significant adrenal tumor, highlighting the need for further characterization with cross-sectional imaging and biochemical correlation to guide safe management and preoperative planning..

Adrenocortical carcinoma

Adrenocortical carcinoma is a rare but aggressive malignancy. Imaging typically shows large, heterogeneous masses with necrosis, hemorrhage, and possible local invasion. Early recognition is essential for appropriate patient management.

Oncocytic adrenal tumors

Oncocytic adrenal tumors are rare cortical neoplasms characterised histologically by abundant mitochondria. Imaging findings are non-specific, and a definitive diagnosis relies on histopathological analysis.

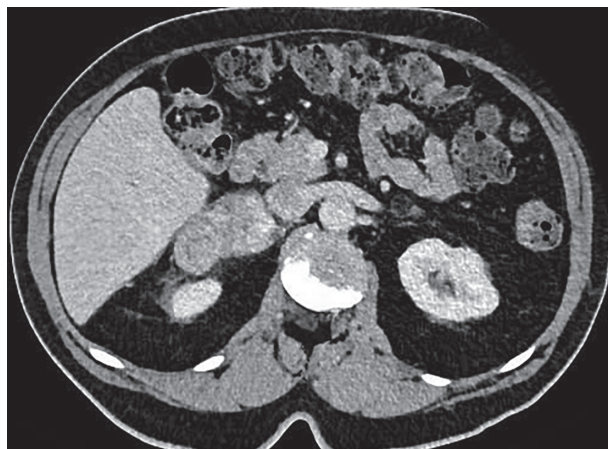


Figure 9. Contrast-enhanced computed tomography of the right adrenal gland demonstrates a well-circumscribed, round mass consistent with a pheochromocytoma. The lesion shows higher unenhanced attenuation compared with typical adenomas and exhibits avid, heterogeneous enhancement following intravenous contrast administration. These imaging features reflect the tumor's hypervascular nature and variable internal architecture, often due to cystic degeneration, necrosis, or hemorrhage. CT evaluation allows accurate assessment of size, morphology, and local invasion, providing essential information for risk stratification, surgical planning, and differentiation from other adrenal masses.

Other rare adrenal lesions

A wide spectrum of additional benign and malignant adrenal lesions has been described, including adrenal cysts, hemorrhage, ganglioneuroma, schwannoma, hemangioma, primary adrenal lymphoma, and rare mesenchymal tumors. These entities are uncommon in routine clinical practice and often lack specific imaging features. As a result, their diagnosis frequently relies on clinical context, lesion evolution over time, and, in selected cases, histopathological confirmation. From a radiological perspective, these lesions are most often categorised as indeterminate, highlighting the intrinsic limitations of imaging in achieving definitive diagnosis for rare adrenal tumors (25,28,30).

Imaging-Based management and follow-up

Imaging plays a central role in determining which adrenal masses require follow-up or intervention.

Lesion size, imaging phenotype, and interval growth are the main parameters considered. Current recommendations emphasise avoiding unnecessary follow-up for lesions with unequivocally benign imaging features, underscoring the responsibility of radiologists in preventing overdiagnosis and overtreatment.

Multidisciplinary implications

The evaluation of adrenal masses requires close multidisciplinary collaboration. Radiologists provide essential information that guides endocrinologists, surgeons, and oncologists in clinical decision-making. Clear reporting, acknowledgment of diagnostic uncertainty, and evidence-based recommendations are key elements of high-quality radiological practice.

Conclusions

Adrenal masses are commonly detected in contemporary imaging practice. A structured, evidence-based, multiparametric imaging approach enables confident lesion characterization and plays a pivotal role in clinical decision-making, risk stratification, and avoidance of unnecessary procedures. Radiologists play a central role in integrating imaging findings into multidisciplinary care pathways.

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